

Daria Krivushkina

Unity Gameplay Systems Engineer

AI-Driven Gameplay · Mobile Performance · C# Architecture

Unity/C# engineer with 4+ years in software and game development. I build deterministic AI-driven gameplay systems, engine-independent C# architecture, reusable gameplay frameworks, and performance-focused mobile systems. My experience covers LLM integration, gameplay architecture, runtime optimization, Android delivery, and production tooling.

Full Technical CV: <https://daridakr.github.io/cv/>

Core Strengths

- **AI Gameplay Architecture** — LLM Function Calling, Narrative Interpreter, deterministic routing, context and memory management.
- **Engine-Independent Architecture** — gameplay and application logic isolated from Unity presentation and external transport.
- **Gameplay Systems Architecture** — reusable runtime abstractions, factories, state machines, ScriptableObject configuration, and typed event contracts.
- **Mobile Performance** — zero-allocation gameplay paths, object pooling, NonAlloc Physics, async workflows, Addressables, CPU and memory profiling.
- **Event-Driven Gameplay** — typed contracts between inventory, production, buildings, quests, AI, and UI.
- **Zero View/Logic coupling** — passive Views with state and behavior owned by Presenters, ViewModels, and gameplay services.

Professional Experience

Unity Developer — DeQuest & Rivalz

November 2023 – December 2025 · Remote, Full-Time

- **AI Gameplay Architecture:** Built an Engine-Independent Architecture that kept LLM transport separate from gameplay state and Unity presentation.
- **AI-Driven Production:** Reduced narrative scripting time by **80%**, from days to hours, by moving dialogue logic into reusable AI-driven gameplay systems.
- **Core Loop Protection:** Ensured **100% core-loop stability against AI hallucinations** through a **Narrative Interpreter** that isolated AI output from direct game-state access and routed actions through explicit gameplay handlers. Implemented **3 Gemini Function Calling handlers** and routed validated actions through explicit gameplay services.
- **Context & Memory:** Built context-selection logic for scene state, quests, inventory, and world flags, with persona memory and bounded conversation history.
- **Async AI Transport:** Kept gameplay responsive during **0.5–7 s AI requests** using UniTask, async/await, and cancellation tokens, with only **2–7 ms measured application overhead**.
- **Frame Stability:** Maintained a **60 FPS main-thread target** while AI requests ran asynchronously and deterministic gameplay processing remained inside Unity systems.
- **Zero-Allocation Gameplay:** Achieved **0 bytes allocated per interaction-scan frame** in the profiled path using Physics.OverlapSphereNonAlloc and cached collider buffers. Used Zenject MemoryPool for repeated runtime objects.
- **Timing System:** Replaced repeated native Update() scheduling with a custom **O(1) Hashed Timing Wheel** for delayed gameplay execution.
- **Gameplay Framework Architecture:** Designed reusable gameplay frameworks and a shared resource pipeline using item contracts, runtime abstractions, factories, state machines, ScriptableObject configuration, and typed **Event-Driven Gameplay**. Kept UI Views passive with Presenter/ViewModel-owned logic for **Zero View/Logic coupling**.
- **Tools & Automation:** Saved **10+ hours of manual configuration per sprint** with custom Editor tools, Android/iOS build profiles, CI/CD automation, and Firebase CLI QA delivery.

Unity Developer — HC Rotem Kuza

August 2023 – November 2023 · Remote, Contract

- **Android Delivery:** Delivered **49 Android APK variants** for QA across game re-skins while maintaining **99.9% build stability** during rapid build cycles.
- **Mobile Monetization:** Integrated ad mediation, rewarded and interstitial ads, IAP, attribution, and analytics SDKs into mobile projects.
- **LeoECS:** Implemented gameplay changes inside an existing LeoECS codebase while working within its entity-component-system architecture.

Unity Developer — Freelance

September 2021 – August 2023 · Remote, Independent Contractor

- Built reusable C# gameplay components with FSM, Observer, and Factory patterns, reused core systems across prototypes, and delivered PC, mobile, and WebGL QA builds.

Selected Projects

Live Farm — AI-Driven Gameplay Systems

2025 · Solo Unity Developer

Reworked the Riftvale concept around deterministic AI integration. Built the Narrative Interpreter, Gemini Function Calling pipeline, context and memory system, asynchronous AI transport, zero-allocation interaction path, and modular gameplay boundaries.

Stack: Unity 2022.3.23f1 LTS · C# · Google Gemini API · LLM Function Calling · Zenject · UniTask · UI Toolkit · EventBus · Addressables · NonAlloc Physics

GitHub: github.com/daridakr/gemini-ai-farm

Riftvale — Mobile AI RPG

2024 · Team Unity Developer, Head Of Development

First playable Android MVP of the AI-driven RPG concept. Established reusable gameplay frameworks for inventory, production, buildings, quests, AI character interactions, state-driven gameplay, and event-driven module communication.

Stack: Unity 2021.3.29f1 · C# · Local LLM API · Zenject · URP · AI Navigation · FSM · ScriptableObjects

GitHub: github.com/daridakr/Riftvale

Forever Village — RPG / WebGL Case

2023 · Solo Unity Developer

Large RPG and village simulation project covering progression, production, construction, economy, combat, equipment, FSM-driven AI, and companion systems. Later used as the basis for a full architecture review and modular redesign.

Stack: Unity 2021.3.29f1 · C# · Zenject · UniTask · URP · FSM · NavMesh · ScriptableObjects · Object Pooling

GitHub: github.com/daridakr/Forever-Village

Education

Diploma in Software Engineering Technician

College of Business & Law

September 2018 – March 2022

Stack

- **Core:** C# · Unity 2021/2022 · OOP · SOLID · Clean Architecture · Engine-Independent Architecture · Dependency Injection
- **AI:** Google Gemini API · LLM Function Calling · Narrative Interpreter · JSON Payload Parsing · Context / State / Memory Management · Deterministic Routing
- **Gameplay Architecture:** Event-Driven Architecture · Zenject · MVP/MVVM · FSM · Strategy · Command/Interpreter · ScriptableObjects
- **Performance:** UniTask · async/await · CancellationToken · NonAlloc Physics · Object Pooling · Zenject MemoryPool · O(1) Hashed Timing Wheel · CPU / Memory Profiling
- **Unity / Mobile:** URP · UI Toolkit · uGUI · Addressables · AssetBundles · New Input System · Android · IL2CPP
- **Production:** Custom Editor Tooling · CI/CD · Firebase CLI · Git · Git LFS · Ad Mediation · IAP · Attribution · Analytics

Contact

Location: Portugal · GMT+1

Email: d.krivushkina@gmail.com

LinkedIn: [linkedin.com/in/daria-krivushkina](https://www.linkedin.com/in/daria-krivushkina)

Full Technical CV: <https://daridakr.github.io/cv/>